{{REPORT_TITLE}}

{{REPORT_SUBTITLE}}

September 15, 2026

Table of contents

Introduction **5**
Acknowledgements 5
Introduction 5

List Of Acronyms **6**

1 {{CHAPTER_TITLES.0}} **7**
Introduction 7
1.1 Introduction to the Project 7
1.2 Presentation of {{ORGANIZATION_NAME}} and the {{PROGRAM_NAME}} Program 7
1.3 Problem Statement 7
1.3.1 Machine reliability vs. generative output 8
1.4 Study of Existing Solutions 8
1.4.1 Category one 8
1.4.2 Category two 8
1.4.3 What is missing 8
1.5 Proposed Solution and Objectives 8
1.5.1 Objectives 8
1.6 Scope & Limitations 8
1.7 Methodology & Approach 9
Conclusion 9

2 Analysis and Specification of Requirements **10**
Introduction 10
2.0.1 Technology and framework choices 10
2.0.2 Testing and tooling 10
Conclusion 10

3 {{CHAPTER_TITLES.2}} 11
Introduction . . . 11
3.1 System Architecture . . . 11
3.2 Domain Modeling . . . 12
3.3 API Design . . . 13
3.4 UI / UX Considerations . . . 13
3.5 Technology Choices . . . 14
3.5.1 Web framework . . . 14
3.5.2 Data and messaging . . . 14
3.5.3 Validation and serialization . . . 14
3.5.4 Messaging and background work . . . 14
3.5.5 Testing . . . 14
3.5.6 Deployment . . . 14
3.6 AI Integration Design . . . 14
3.6.1 The unified client . . . 14
3.6.2 Taming unstructured output . . . 14
3.6.3 Provider mapping and fallback . . . 14
3.6.4 Hybrid pipeline design . . . 14
Conclusion . . . 14

4 {{CHAPTER_TITLES.3}} 15
Introduction . . . 15
4.1 Development Environment & Tools . . . 15
4.2 Implementation of Core Features . . . 15
4.2.1 Module 1 . . . 15
4.2.2 Module 2 . . . 15
4.2.3 Module 3 . . . 15
4.2.4 Evaluation and confidence . . . 15
4.3 Results & Impact . . . 16
4.4 Future Work . . . 16
4.4.1 Feature X . . . 16
4.4.2 Refinements . . . 16
4.4.3 Hardening . . . 16
Conclusion . . . 16

General Conclusion 17

References 18

Appendices **19**

A Code **19**
A.1 Key algorithm: one important piece of code 19
A.2 Key algorithm: another important piece of code 19
A.3 Worker: background processing 19

Platform Screenshots **20**
Module A 20
Module B 21
Module C 22

Introduction

Acknowledgements

{{ACKNOWLEDGEMENT_TEXT}}

Introduction

This internship report presents the project **{{PROJECT_NAME_DISPLAY}}**, carried out at {{ORGANIZATION_NAME}} as part of the {{PROGRAM_NAME}} program, during the period from {{INTERNSHIP_START}} to {{INTERNSHIP_END}}.

The report is structured into four chapters:

- **Chapter 1, {{CHAPTER_TITLES.0}}** (Chapter 1) — gives the context of the project and the motivation.
- **Chapter 2, {{CHAPTER_TITLES.1}}** (Chapter 2) — analyzes the requirements.
- **Chapter 3, {{CHAPTER_TITLES.2}}** (Chapter 3) — presents the design.
- **Chapter 4, {{CHAPTER_TITLES.3}}** (Chapter 4) — details the implementation and results.

List Of Acronyms

1 {{CHAPTER_TITLES.0}}

Introduction

This chapter presents the context of the internship and the project: the organization, the program, the problem addressed, existing solutions, the proposed solution and its objectives, the scope, and the methodology used.

1.1 Introduction to the Project

This internship took place at {{ORGANIZATION_NAME}} from {{INTERNSHIP_START}} to {{INTERNSHIP_END}}. The project, {{PROJECT_NAME_DISPLAY}}, addresses the following problem: {{PROJECT_DESCRIPTION}}.

1.2 Presentation of {{ORGANIZATION_NAME}} and the {{PROGRAM_NAME}} Program

{{ORGANIZATION_DESC}}

1.3 Problem Statement

The internship started from the following difficulties:
{{PROBLEM_STATEMENT}}

1.3.1 Machine reliability vs. generative output

1.4 Study of Existing Solutions

1.4.1 Category one

- *Solution A* (short description)
- *Solution B* (short description)

1.4.2 Category two

- *Solution C*
- *Solution D*

1.4.3 What is missing

1.5 Proposed Solution and Objectives

The proposed solution, {{PROJECT_NAME_DISPLAY}}, addresses the gaps above by delivering:
{{SOLUTION_BULLETS}}

1.5.1 Objectives

The main objectives of the project are to:

- Objective 1: …
- Objective 2: …
- Objective 3: …
- Objective 4: …

1.6 Scope & Limitations

The project scope covers the following modules (in priority order):

- **Core module 1** — one-line description.
- **Core module 2** — one-line description.
- **Supporting module** — one-line description.

The main limitations are:

- {{PROBLEM1}}
- {{PROBLEM2}}

1.7 Methodology & Approach

The project was carried out in {{INTERNSHIP_DURATION}} iterative sprints, following this structure:

Table 1.1: Methodology

Sprint	Duration	Deliverables
Sprint 1	Week 1-2	{{DELIVERABLE_1}}
Sprint 2	Week 3-4	{{DELIVERABLE_2}}
Sprint 3+	…	{{DELIVERABLE_3}}

Development used a {{REPO_HOST}} repository ({{REPO_PUBLIC_URL}}), merge requests gated by a {{CI_SYSTEM}} pipeline, and {{TEST_FRAMEWORK}} for automated tests.

Conclusion

This chapter introduced the internship context, the problem, the proposed solution, and the methodology. The next chapter analyses the requirements.

2 Analysis and Specification of Requirements

Introduction

This chapter analyzes the requirements for the {{PROJECT_NAME}} platform described in the previous chapter. It starts with the stakeholders, then derives the principles and standards that shape every decision, and finally captures the functional and non-functional requirements.

2.0.1 Technology and framework choices

{{TECH_CHOICES_CONTENT}}

2.0.2 Testing and tooling

{{TESTING_CONTENT}}

Conclusion

This chapter analyzed the requirements and constraints of {{PROJECT_NAME}}. The following chapter translates these into an architecture and a detailed design.

3 {{CHAPTER_TITLES.2}}

Introduction

This chapter presents the design of {{PROJECT_NAME_DISPLAY}}: the system architecture, the domain model, the API design, the UI/UX considerations, the technology choices, and the AI integration design.

3.1 System Architecture

{{PROJECT_DESCRIPTION}} The architecture follows a modular design where each business capability is an independent named module.

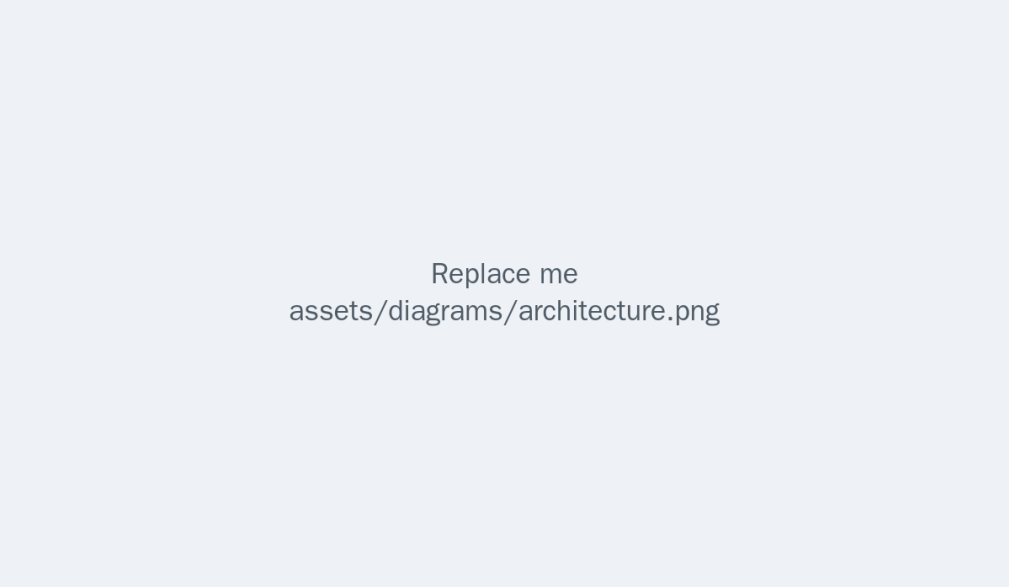


Figure 3.1: System architecture

3.2 Domain Modeling

The core domain entities are modeled with {{VALIDATION_LIBRARY}} (e.g., Pydantic or dataclasses), ensuring structured validation and one source of truth across the API and the workers. The central entity of the domain is the PROJECT_ENTITY, modeled as follows:

```
class AtomicTask(BaseModel):
    id: str
    title: str
    description: str
```

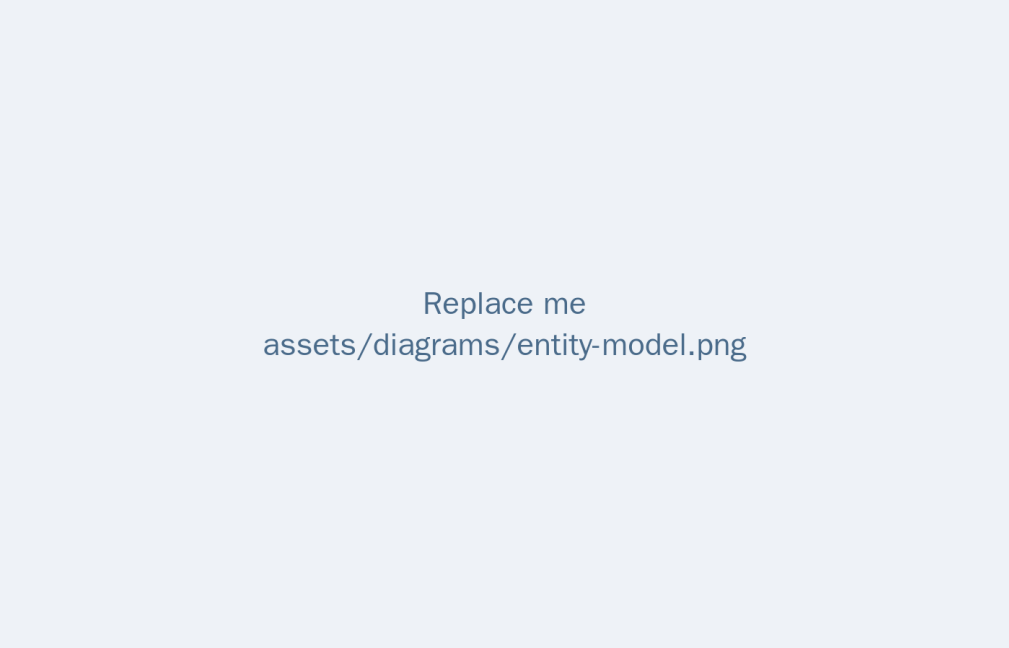


Figure 3.2: Entity-relationship model

3.3 API Design

The platform exposes a RESTful API (HTTP) grouped by module. The main endpoints are summarized below.

Table 3.1: Main API endpoints

{{MODULE}}	{{METHOD}}	{{PATH}}	{{DESCRIPTION}}
Module 1	GET	`/api/v1/module1`	…
Module 1	POST	`/api/v1/module1`	…
Module 1	GET	`/api/v1/module1/<id>`	…
Module 2	POST	`/api/v1/module2`	…
Module 2	GET	`/api/v1/module2/<id>`	…
…	…	…	…

3.4 UI / UX Considerations

The platform's primary interface is its API, and the user experience is shaped through response design rather than screens. The main decisions were:

- **Decision 1** — one line.
- **Decision 2** — one line.
- **Decision 3** — one line.

3.5 Technology Choices

3.5.1 Web framework

3.5.2 Data and messaging

3.5.3 Validation and serialization

3.5.4 Messaging and background work

3.5.5 Testing

3.5.6 Deployment

3.6 AI Integration Design

3.6.1 The unified client

3.6.2 Taming unstructured output

3.6.3 Provider mapping and fallback

Table 3.2: Capability-to-model mapping

{{CAPABILITY}}	{{MODEL_ALIAS}}	{{PROVIDER}} / {{MODEL}}	{{USE_CASE}}
`capability_1`	`model-1`	Provider `model-name`	…
`capability_2`	`model-2`	Provider `model-name`	…

3.6.4 Hybrid pipeline design

Conclusion

This chapter presented the design of {{PROJECT_NAME_DISPLAY}}. The next chapter describes its implementation.

4 {{CHAPTER_TITLES.3}}

Introduction

This chapter details the implementation of {{PROJECT_NAME_DISPLAY}}. It begins by presenting the development environment, then explains, module by module, how each feature was implemented. After that it examines the results achieved and outlines possible future work.

4.1 Development Environment & Tools

The project repository, `{REPO_NAME}`, is hosted on {{REPO_HOST}} at {{REPO_URL}}. It is a {{LANGUAGE}} project. {{DEPLOYMENT_METHOD}} orchestrates the stack:

4.2 Implementation of Core Features

4.2.1 Module 1

4.2.2 Module 2

4.2.3 Module 3

4.2.4 Evaluation and confidence

Table 4.1: Confidence guidelines

{{FIELD}}	{{TYPICAL_CONFIDENCY}}	{{NOTES}}
`short_field`	~90-95%	—
`longer_field_name`	~85%	—

{{FIELD}}	{{TYPICAL_CONFIDENCE}}	{{NOTE}}

4.3 Results & Impact

By the end of the internship, {{PROJECT_NAME_DISPLAY}} had a complete core:

4.4 Future Work

4.4.1 Feature X

4.4.2 Refinements

4.4.3 Hardening

Conclusion

This chapter described the implementation of {{PROJECT_NAME_DISPLAY}}, the results achieved, and future improvements.

General Conclusion

This {{INTERNSHIP_DURATION}} internship, focused on developing {{PROJECT_NAME_DISPLAY}}, …

It reinforced a mindset that will benefit any future software engineering work: software expands what a team can build, but engineering discipline decides whether what is built can be trusted.

References

A Code

A.1 Key algorithm: one important piece of code

```
def example_function(name: str) -> str:
    return f"Hello, {name}!"
```

A.2 Key algorithm: another important piece of code

```
def another_example(items):
    return sorted(items, key=lambda x: x.priority)
```

A.3 Worker: background processing

```
while True:
    _, payload = queue.blpop("job_queue", timeout=30)
    if not payload:
        continue
    result = worker(payload)
    key = f"result:{payload.correlation_id}"
    store.set(key, result)
```

Platform Screenshots

Module A

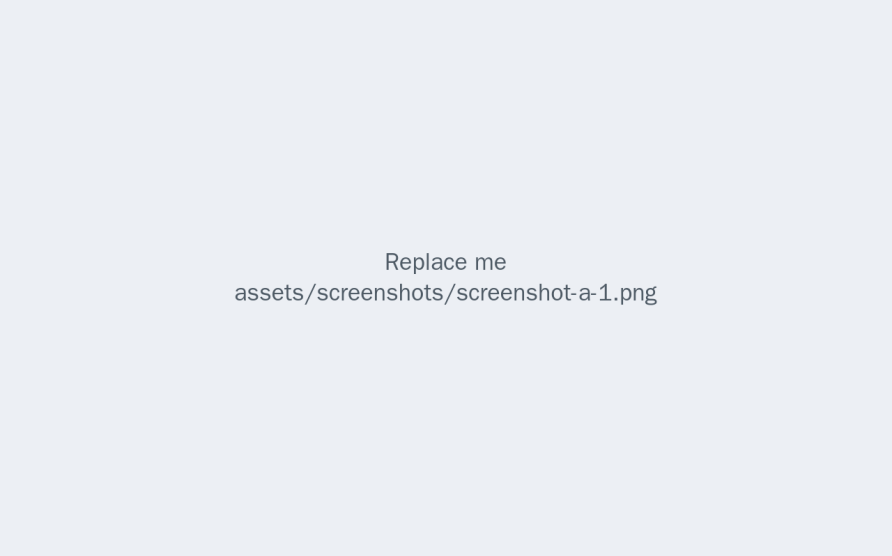


Figure A.1: Caption describing what this screenshot shows

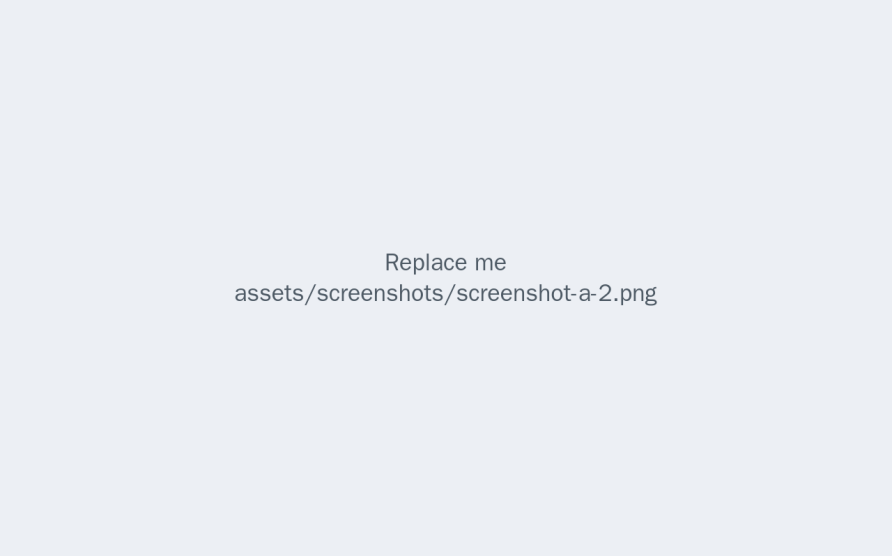


Figure A.2: Second view

Module B

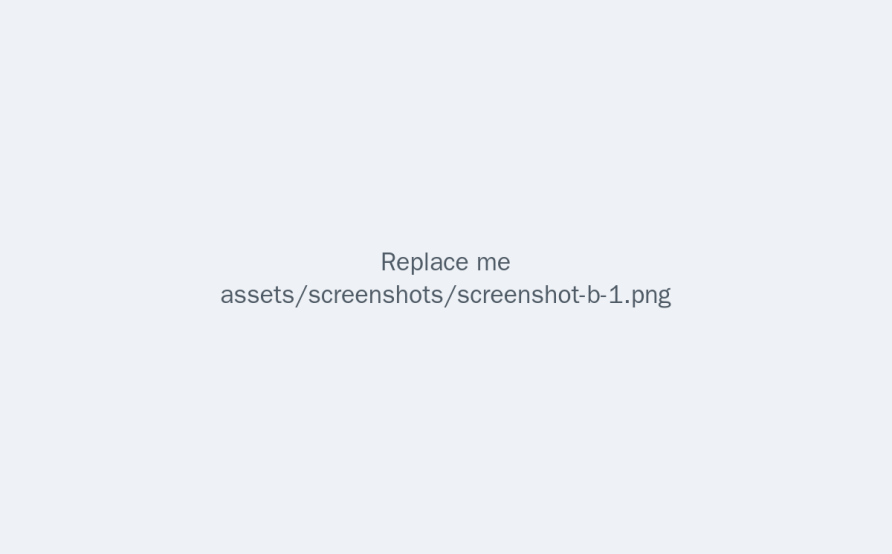


Figure A.3: Caption for module B

Module C

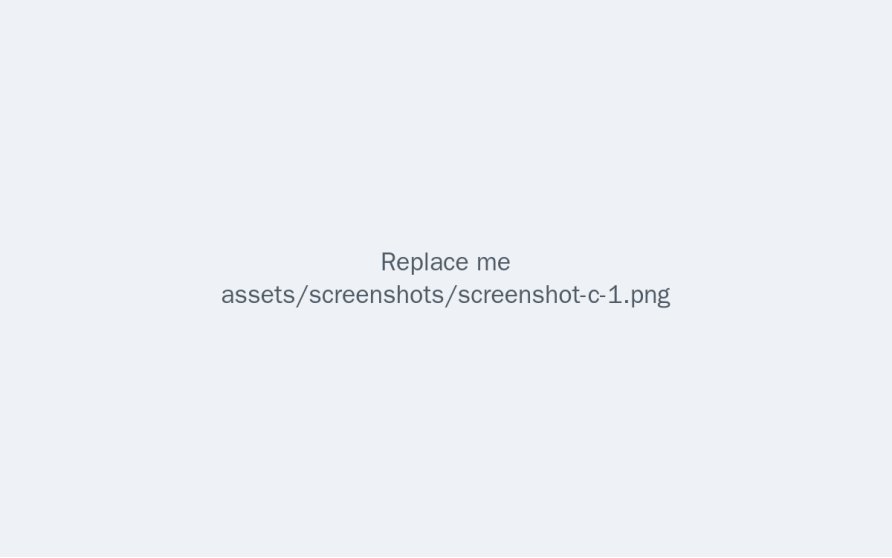


Figure A.4: Caption for module C